

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2– Manipulation (BL3)	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)		

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 Section / Batch: 2024 / Slot D Date: 01/08/26.

Code of Conduct

/ Nagendra (192472399) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nagendra Reddy

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Image Enhancement in Spatial Domain	Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques	Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering	Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain	Enhancing images using frequency domain transformations	4	10
Image Segmentation	Partitioning images into meaningful regions	5	10
GRAND TOTAL:			50 Marks

1. Explain the concept of image enhancement in the spatial domain. Discuss how basic gray-level transformations, histogram processing techniques, and spatial filtering can be used to improve the quality of the captured images. Justify the suitability of each technique for this scenario.
2. Differentiate between image smoothing and image sharpening. Explain the working principles of spatial smoothing filters and sharpening filters. Discuss how these techniques improve image quality and explain the situations in which each technique should be applied.
3. Explain the image enhancement process in the frequency domain. Describe the roles of low-pass filters and high-pass filters in image enhancement. Compare their effects on image quality and discuss suitable applications for each filtering technique.
4. Explain the concept of image segmentation. Compare point detection, line detection, edge detection, thresholding, and region-based segmentation with respect to their working principles, advantages, limitations, and practical applications. Identify the most appropriate technique(s) for this scenario and justify your answer.
5. Explain the importance of edge detection, edge linking, and boundary detection in image processing. Discuss the working principles of these techniques and describe how they contribute to accurate object recognition and defect detection. Illustrate your answer with suitable examples and diagrams wherever necessary.

Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Faculty In-charge

Course Coordinator

Head of Department

Signature: _____

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Definition :- Image enhancement in the spatial domain improves the visual quality of an image by directly modifying the pixel values.

The basic transformation is :

$$g(x, y) = T[f(x, y)]$$

a) Gray-level Transformations : It Modify the intensity values of pixel

Types :

- Image negative
- Log Transformation
- Power law

Advantages :

- Improves brightness
- Enhances contrast

Applications :

- Medical images
- Satellite images.

b) Histogram Processing : It represents the distribution of gray levels.

Techniques :

- Histogram Equalization
- Histogram stretching

Advantages :

- Improves global contrast

- Enhances dark and bright regions.

Applications :

- X-ray images
- Remote Sensing

c) Spatial Filtering :- It operates on a neighborhood of pixels using a kernel.

Smoothing Filters :

- Mean
- Median
- Gaussian

Advantages :

- Enhance edges
- Improve object boundaries.

Justification

Technique	Purpose	Suitable for
Gray-level Transformation	Brightness	Low-light images
Histogram Processing	Contrast enhancement	Dark-low contrast image
Spatial filtering	Noise removal	Noisy and blurred images.

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③ Definition:- Frequency-domain enhancement modifies the image after converting it into the frequency domain using Fourier-Transform.

Steps:

Image $\rightarrow$ Fourier Transform $\rightarrow$ Filtering $\rightarrow$ Inverse Fourier transform $\rightarrow$ Enhanced image

Low-Pass filter

Principle:- Allows low frequency components and removes high-frequency components.

Effects:-

- Smooth image
- Remove noise

Applications for High-Pass Filter.

- Face recognition
- Fingerprint enhancement

High-Pass filter:-

Principle:- Allows high frequency components and suppresses and low-frequency components.

Effects:-

- Sharpens image
- Enhances edge

Image after
form -

Definition :- Image Segmentation is the Process of dividing an image into meaningful regions or objects for easier analysis.

Technique	Working principle	Application
Point detection	Detects isolated pixels	Defect detection
Line detection	Detects horizontal, vertical or diagonal lines	Road and crack detection
Edge detection	Detects sudden intensity changes	Object boundary detection
Thresholding	Separates object based on pixel intensity	Document and medical images
Region-Based Segmentation	Group similar neighboring pixels	Medical image analysis

Most Suitable Technique :-

Edge detection is best for identifying object boundaries while thresholding is suitable when the foreground and background have clear intensity differences.

⑤ a) Edge detection

Definition :- Detects sudden intensity changes that represent object boundaries.

Methods :- Sobel, Prewitt, Canny.

Applications :- Face recognition, Medical imaging.

b) Edge Linking

Definition :- Connects broken edge pixels to form continuous object boundaries.

Methods :- Hough Transform, Graph-based methods.

Advantages :- Produces continuous edge, Improve shape detection.

c) Boundary Detection

Definition :- Extracts the outer boundary of segmented objects.

Methods :- Contour tracing, Chain codes.

Applications :- Pattern recognition, Shape analysis.

Example :-

- Edge detection identified cracks on a metal surface.

- Edge linking connects broken crack segments.

- Boundary detection outlines the crack, allowing accurate measurement of its size and location.